Session 1. Categorical variables.

Pre course: Statistics.

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Agenda

- Structured data bases.
- Types of variables (e1).
- Frequency tables (e2).
- Mean (e2).
- Visualizations (e2 and e3).
- Exercises.

Agenda

- **Structured data bases.**
- Types of variables (e1).
- Frequency tables (e2).
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- Exercises.

Data bases.

- We will use structured data bases only, but there are others out there too.
- Rows represent units (persons, households, towns, countries, dates, whatever is you unit of analysis).
- Columns represent variables (which take values for each of your units).
- There can be different kinds of variables in one data set.

Analisis estadistico - RStudio Source Editor

df_dta

Filter | Cols: « < 1 - 50 > »

	id_vivienda Identificación vivienda	folio Identificación hogar (id_vivienda hogar)	id_persona Identificador de la persona en el hogar	region Región	area Área	cod_upm Código Unidad Primaria de Muestreo (UPM)	nse Nivel Socioeconómico de la UPM
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14	1000906	100090601	1	16	2	10009	4
15	1000906	100090601	2	16	2	10009	4
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17	1000906	100090601	4	16	2	10009	4
18	1000907	100090701	1	16	2	10009	4
19	1000907	100090701	2	16	2	10009	4
20	1000908	100090801	1	16	2	10009	4
21	1000908	100090801	2	16	2	10009	4
22	1000909	100090901	1	16	2	10009	4
23	1000910	100091001	1	16	2	10009	4
24	1001301	100130101	1	8	2	10013	6
25	1001301	100130101	2	8	2	10013	6
26	1001302	100130201	1	8	2	10013	6
27	1001302	100130201	2	8	2	10013	6
28	1001303	100130301	1	8	2	10013	6
29	1001305	100130501	1	8	2	10013	6
30	1001305	100130501	2	8	2	10013	6
31	1001305	100130501	3	8	2	10013	6

Showing 13 to 31 of 202,231 entries, 918 total columns

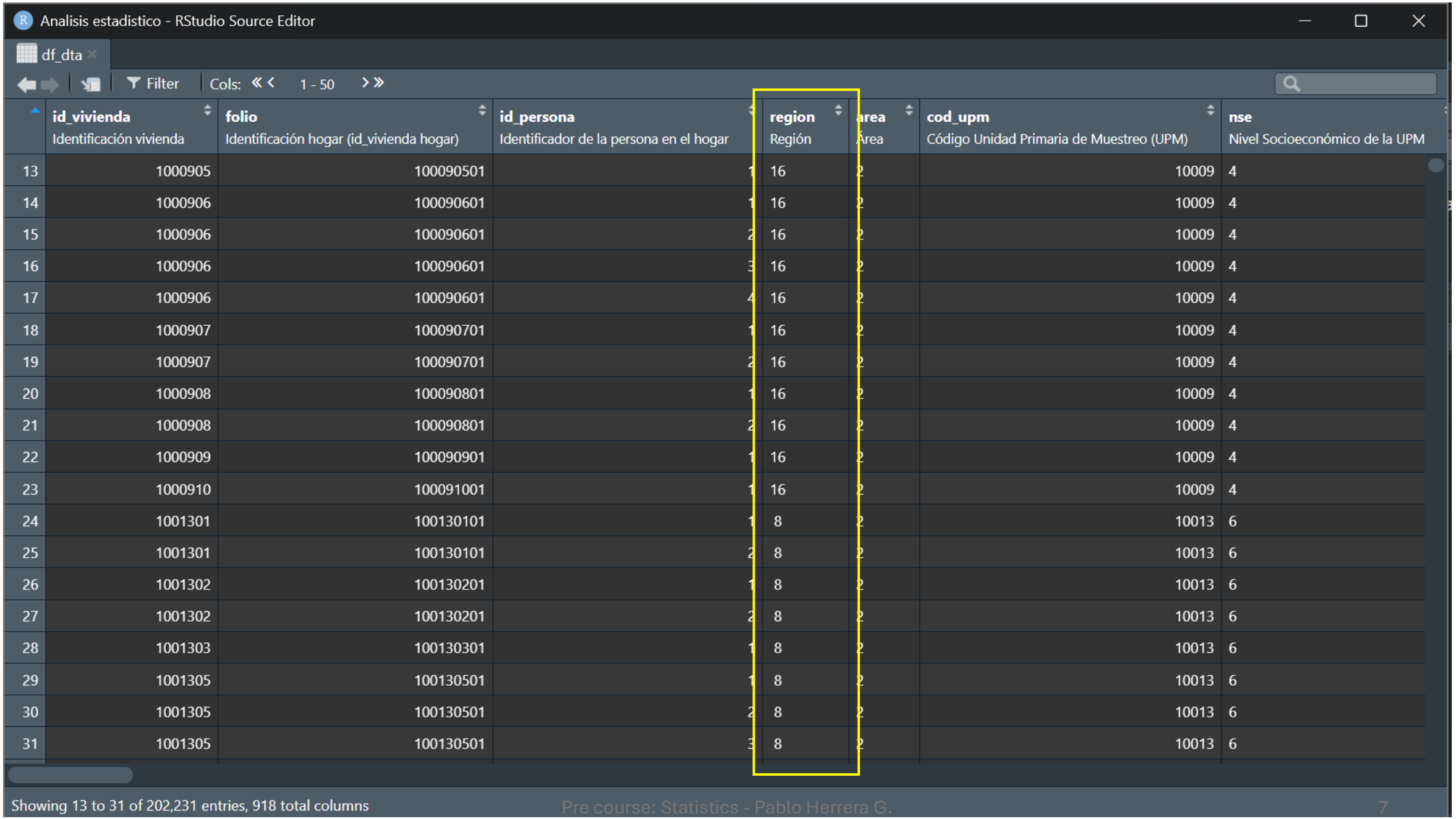
Analisis estadistico - RStudio Source Editor

df_dta

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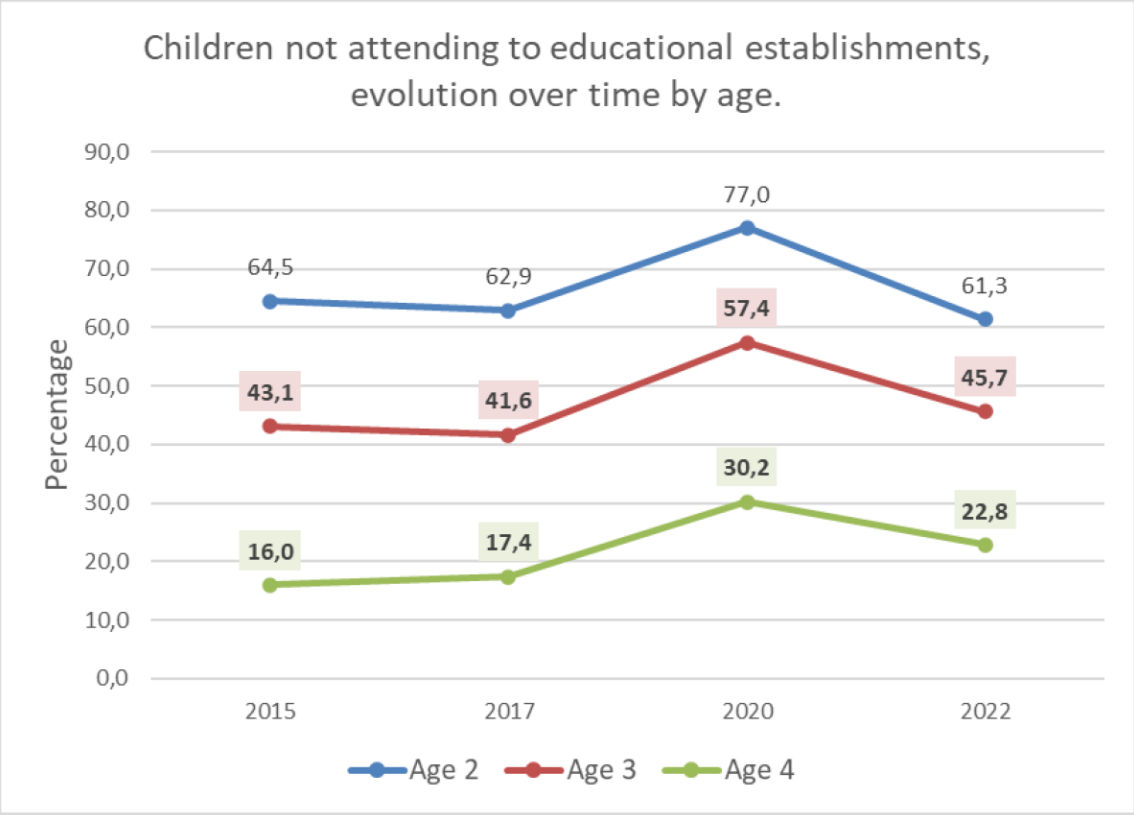
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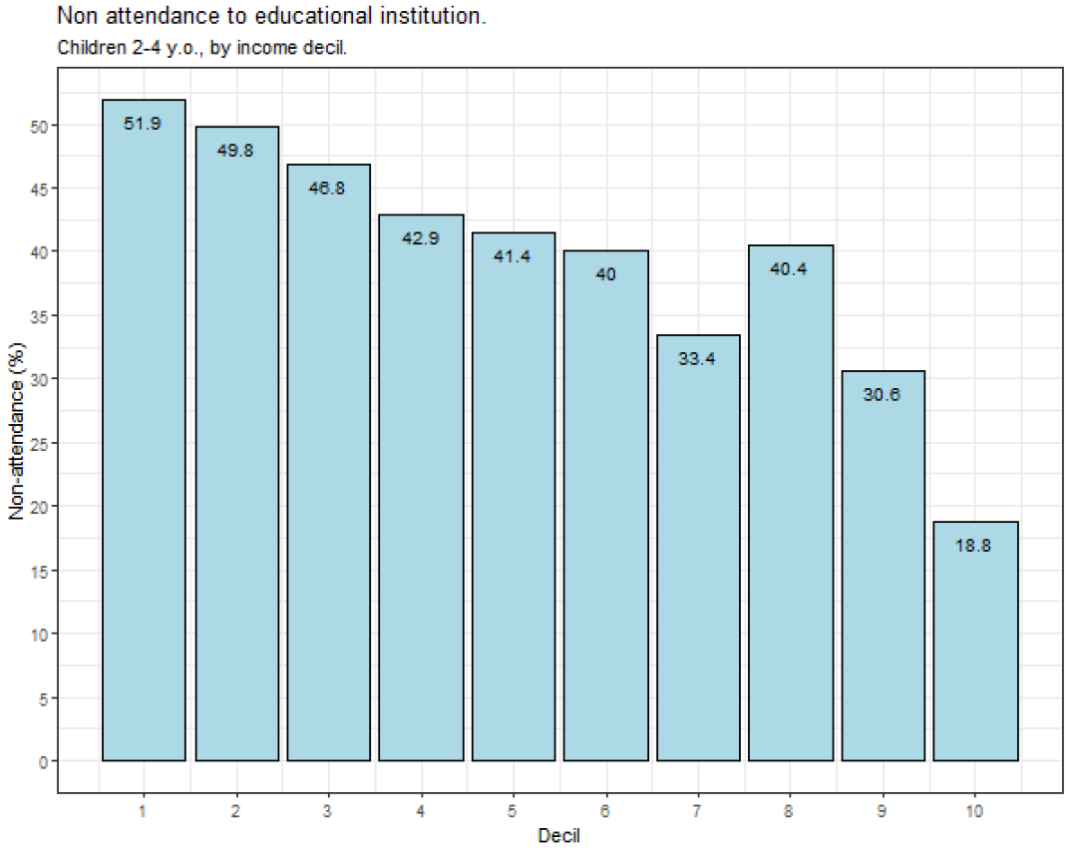
Showing 13 to 31 of 202,231 entries, 918 total columns


Analisis estadistico - RStudio Source Editor
df_dta
Filter
Cols:
1 - 50
id_vivienda
Identificación vivienda
folio
Identificación hogar (id_vivienda hogar)
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Identificador de la persona en el hogar
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Data bases.

- They can help us to do useful analysis...





Data bases.

- Rows represent units.
- Columns represent variables.
- There can be different kinds of variables in one data set.

- But a data set does not need to have thousands of values to be one.

Data bases.

Household id	Gender head	Employment head	Poverty situation	N working age	N non working age
1	Female	Unemployed	Extreme poverty	1	3
2	Female	Informal	Poverty	2	2
3	Female	Informal	No poverty	2	4
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- Structured data bases.
- **Types of variables (e1).**
- Frequency tables (e2).
- Mean (e2).
- Visualizations (e2 and e3).
- Exercises.

Types of variables.

- Categorical variables (qualitative).
  - Ordinal variables (order does matter).
  - Nominal variables (order **does not** matter).
    - Binary (or dummy or 0,1) included.

- Discrete variables.

- Continuous variables.

Types of variables.

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id	dummy	categorical	categorical	discrete	discrete

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Frequency tables.

- Used to summarize the distribution of a population or sample
- Population separated according to a categorical variable.
- Population (or Frequency or N), Total, Percentage, Cumulative percentage.

Dependency level	Population	%	Cumulative %
No dependency	13.856.983	90,2	90,2
Mild dependency	529.274	3,4	93,7
Moderated dependency	549.424	3,6	97,3
Severe dependency	420.279	2,7	100,0
Total	**15.355.960**	**100**	

Frequency tables.

- We could create one with any categorical variable of our data set:

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Mean.

- Also known as the average.
- Is the most common centrality measure.
- There is also the median and the mode (session 5).
- It can be used with numerical variables only (both discrete and continuous).
- Avoid it when using categorical variables.

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

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Mean of selected column =

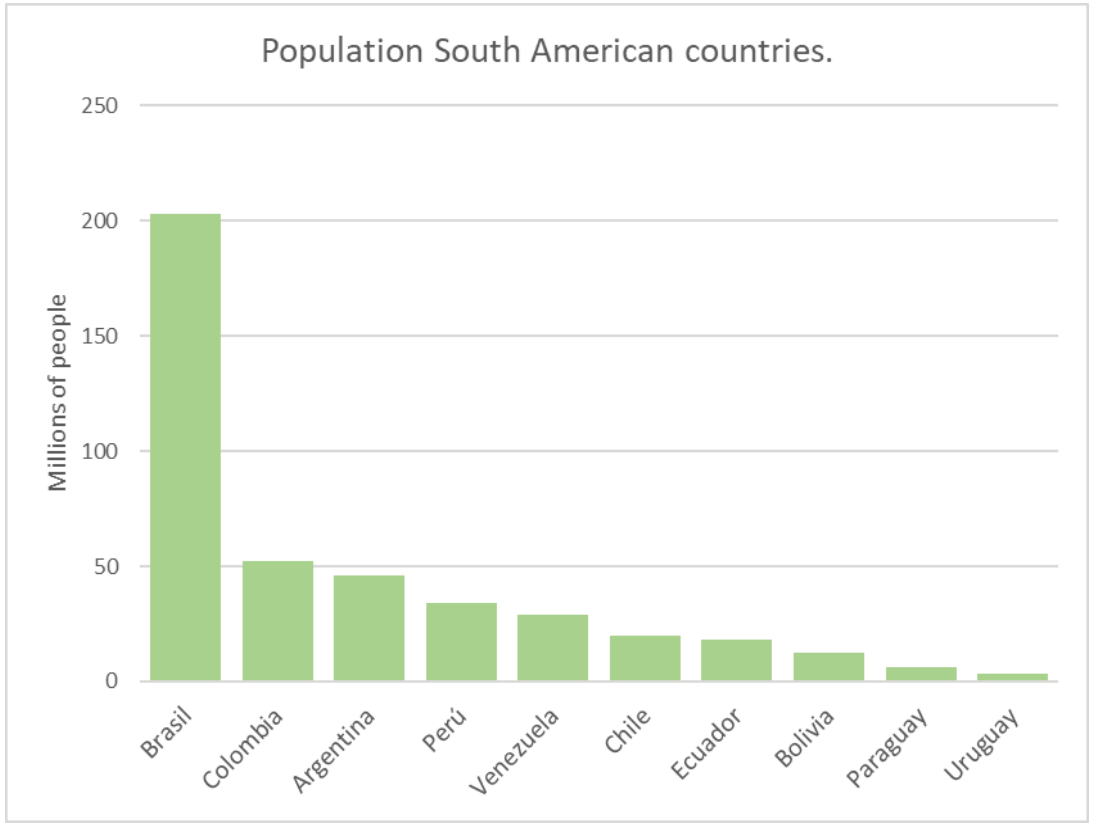
(1 + 2 + 2 + 1 + 2 + 1 + 2 + 3 + 3 + 2) / 10 = 1.9

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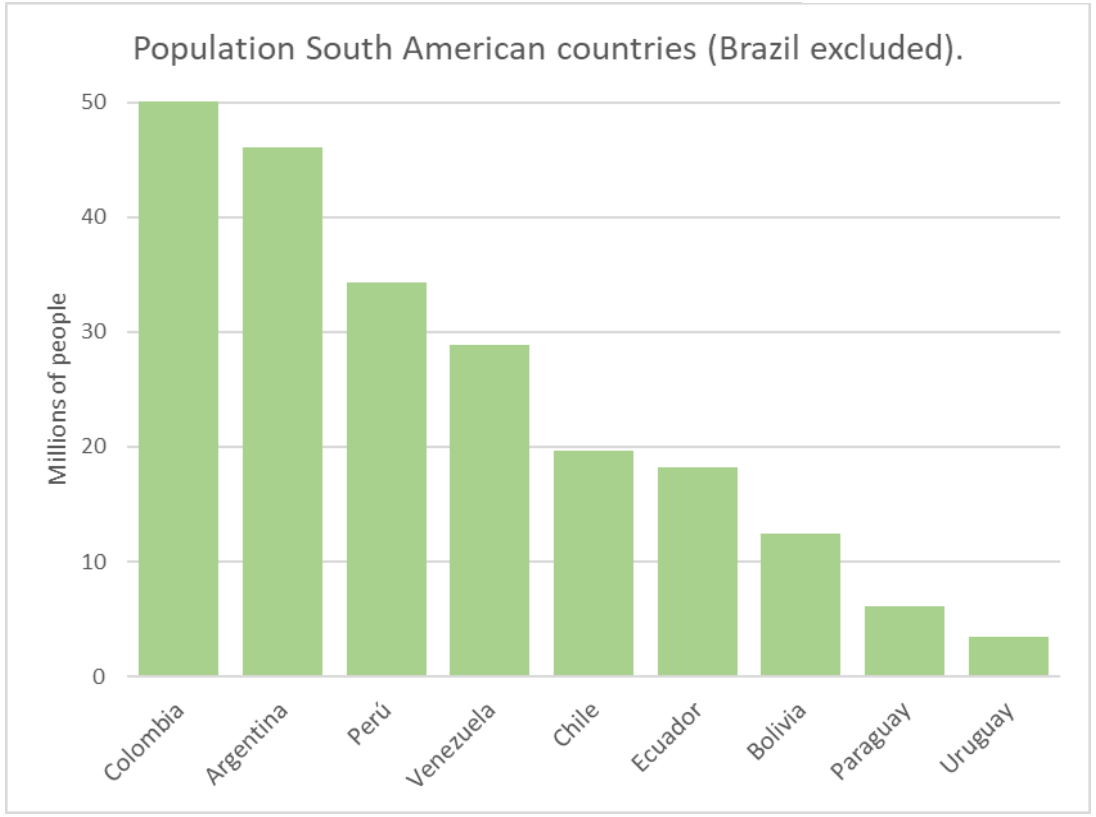
Visualizations.

- For this session, we will talk only about bar plots.
- In general, data visualizations:
  - Can be helpful to make a point clear.
  - Can summarize a lot of information at the same time.



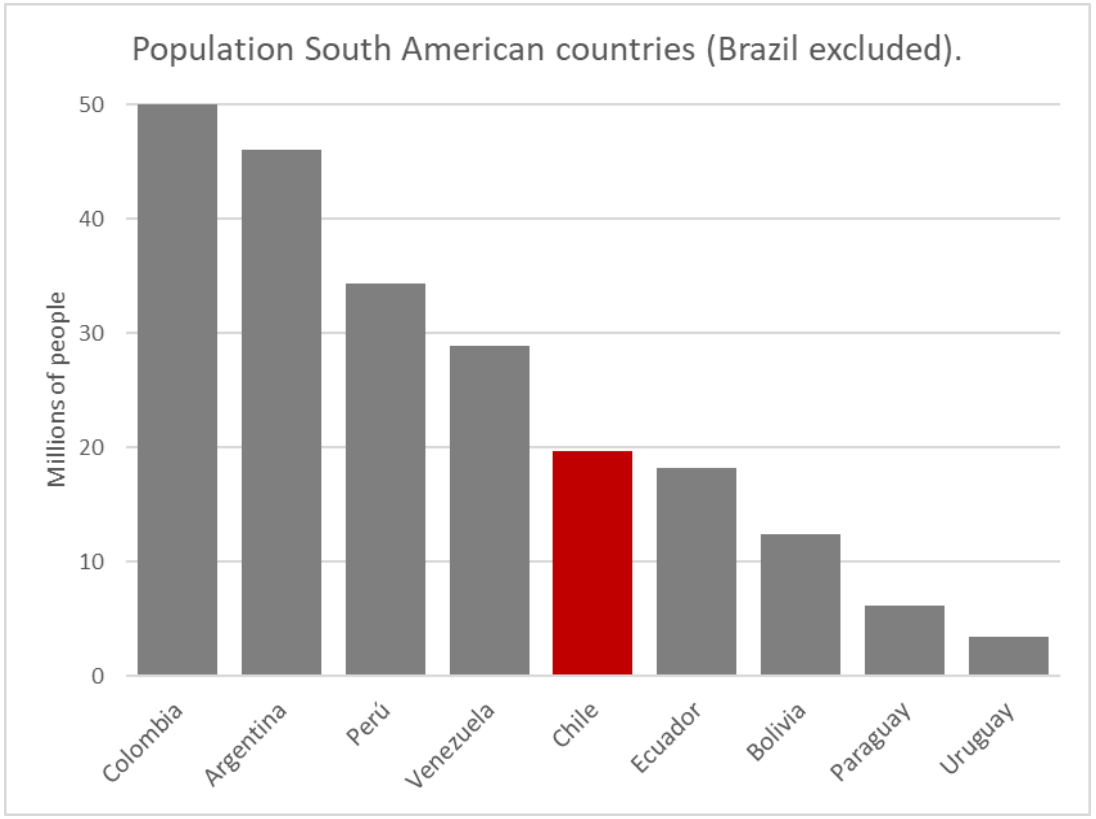
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- For this session, we will talk only about bar plots.
- In general, data visualizations:
  - Can be helpful to make a point clear.
  - Can summarize a lot of information at the same time.
  - We can adjust them.
  - Also highlight relevant ideas.



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- **Exercises.**

Exercise 2.

a) Identify types of variables (considering the kind of variables used in the Ex. 1).

b) Present the following:

   i. Bar plot of the employment status.

   ii. Frequency table of poverty situation (number of observations, percentage and cumulative percentage).

   iii. Estimate the mean of the total number of persons by household

c) Present in bullet points the three most relevant take aways from your analysis.

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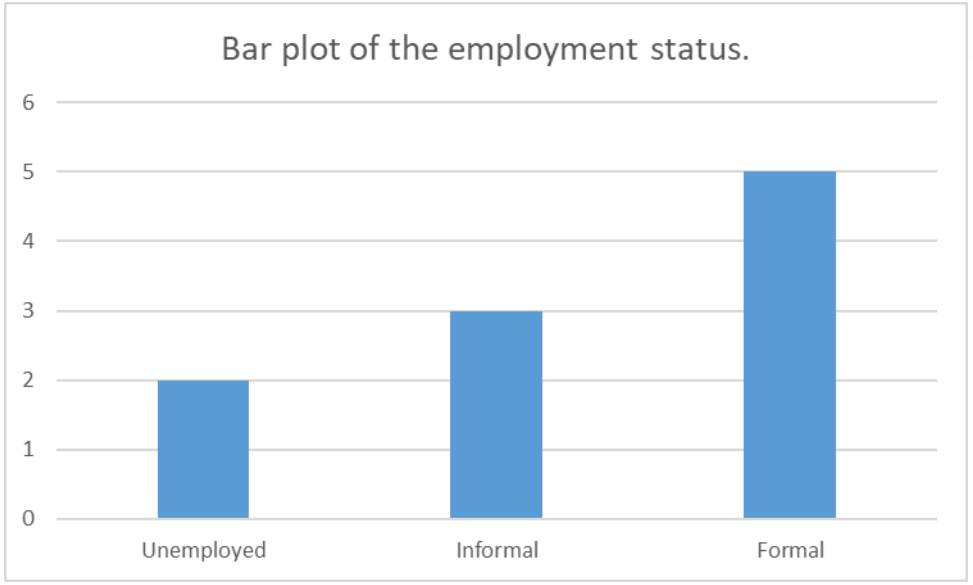
Exercise 2.

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i. Bar plot of the employment status.

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iii. Estimate the mean of the total number of persons by household



Poverty situation	N	%	Cumulative %
Extreme poverty	3	30	30
Poverty	3	30	60
No poverty	4	40	100
Total	**10**	**100**	

Mean of the total number of persons by household

= (N working age + N non working age) / N of households
= (4 + 4 + 6 + 4 + 4 + 4 + 5 + 7 + 8 + 6) / 10
= 5.2

Exercise 2.

c) Present in bullet points the three most relevant take aways from your analysis.

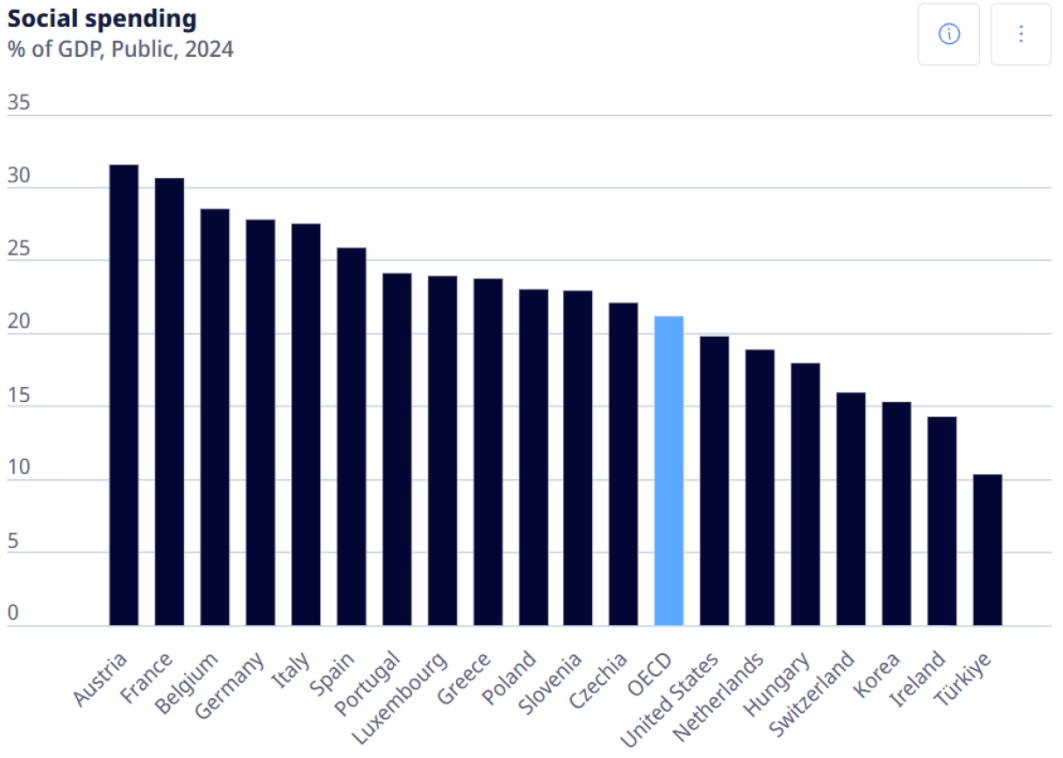
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Exercise 3.

a) What does the bar plot shows? What conclusions general ideas could you extract from it? (a good starting point could be identifying the kind of plot used, the variable that it displays, the year of the data, source of the information). If you want to go further, identify one or more groups of countries to mention something that you would like to highlight.

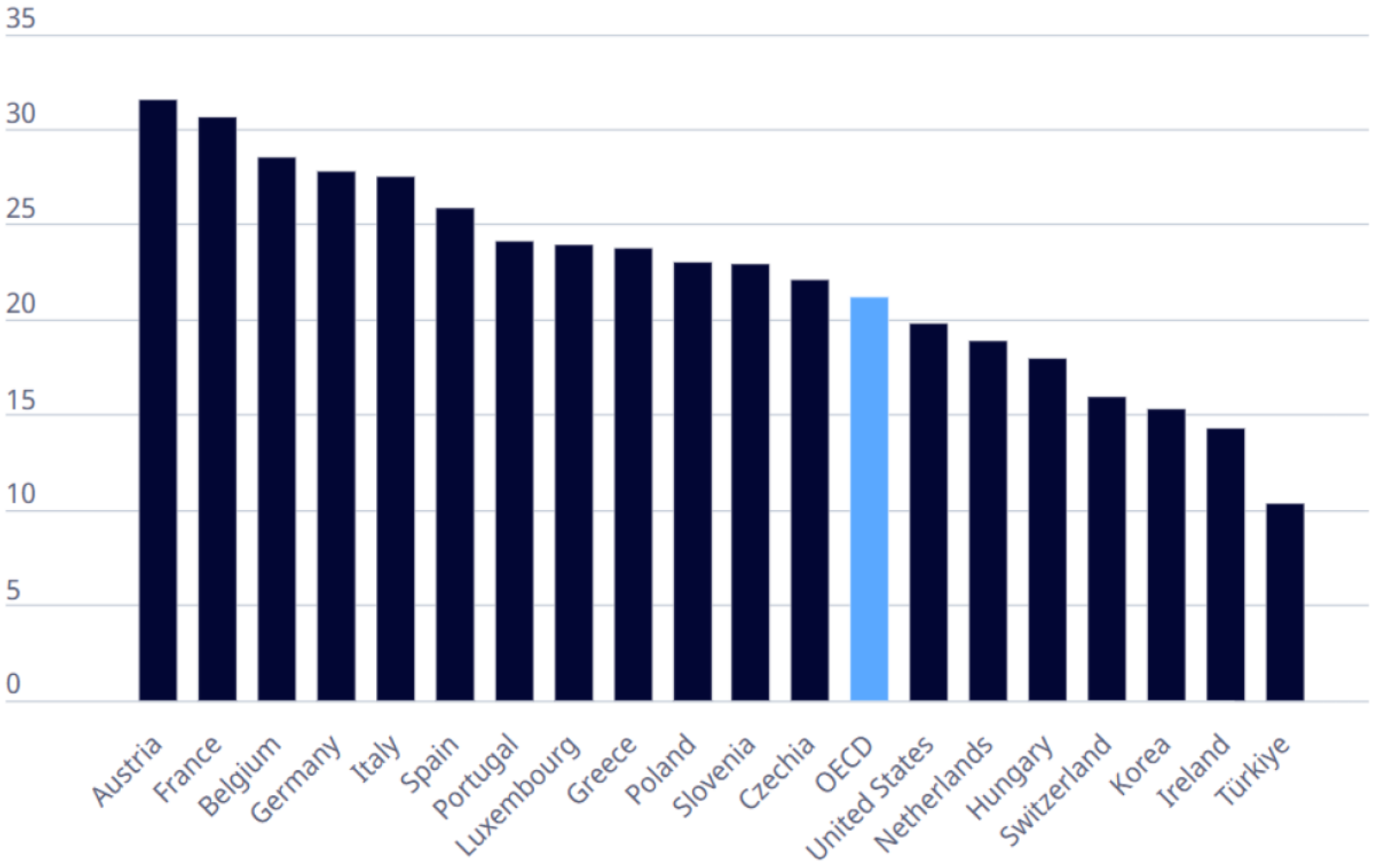
b) Considering the countries displayed in the bar plot, how many of them spend more than 20% of their GDP on social purposes?

c) Considering the countries displayed in the bar plot, which is the one with the lowest social spending? How much is that?



Source: OECD Data. Available at: https://www.oecd.org/en/data/indicators/social-spending.html

Social spending
% of GDP, Public, 2024



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